

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem
Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2820–2826; original printed p. 94; physical source-PDF p. 82; recomposed running p. 74. The printed-page-94 marker occurs before this unit at line 2813. Blank line 2827 is excluded; line 2828 begins the $n = 1$ case and is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration.

We shall prove that (iv) implies (ii). To this end, recall (ii):

(ii) For every $x \in U$ such that $c(x) = 1$, one has

$$\text{prof } \mathcal{F}_x \geq n.$$

We argue by induction on n .

If $n = 0$, both conditions are vacuous.